Organic Farming Fundamentals

Test document 1 of 5 · Concepts, principles, and farm-system design

Purpose: This PDF is an original, concise test document designed to exercise multi-PDF retrieval and page-level citations. It summarizes general concepts and points to authoritative public references; it is not a replacement for official standards or local agricultural advice.

1. What is organic farming?

Organic farming is a production-management approach that emphasizes ecosystem health, biological cycles, biodiversity, and soil biological activity. It generally avoids synthetic fertilizers and pesticides and instead relies on practices such as crop rotation, compost, green manures, biological pest management, and locally adapted farming methods.

Organic is not simply a label for food that looks natural. Where a product is marketed as certified organic, the producer and supply chain must follow the applicable standards and certification or verification requirements.

2. Core principles

Build and maintain soil fertility over the long term rather than treating the crop as an isolated production unit.

Recycle plant residues and suitable animal-derived materials so nutrients return to the soil.

Support biodiversity, including beneficial insects, pollinators, soil organisms, and diverse crop rotations.

Use preventive cultural, biological, and mechanical practices to manage weeds, pests, and diseases.

Protect water, soil, and air from avoidable pollution and adapt methods to local climate and farm conditions.

3. Planning a transition

Before changing systems, document field history, soil condition, current inputs, crop rotations, irrigation, pest pressure, and available organic materials. The required conversion period depends on the applicable standard and farm circumstances.

Keep records of seed and planting material, field operations, purchased inputs, compost preparation, harvests, storage, and sales. Check the relevant national organic standard before making certification claims.

4. Key takeaway

Organic farming is a whole-farm management system. Individual techniques such as composting or biological pest control are most effective when combined with soil management, crop diversity, careful monitoring, and credible records.

Questions to test your RAG application

1. What are four core principles of organic farming?
2. Why is organic farming considered a whole-farm management system?
3. What should a farmer document before transitioning to organic production?

Public references

FAO — Organic Agriculture
https://www.fao.org/platforms/green-agriculture/areas-of-work/natural-resources-biodiversity-green-production/organic-agriculture/

FAO/WHO — Guidelines for the Production, Processing, Labelling and Marketing of Organically Produced Foods
https://www.fao.org/4/w9087e/w9087e0i.htm

Tamil Nadu Agricultural University — Organic Farming portal
https://www.agritech.tnau.ac.in/org_farm/orgfarm_index.html

Source note: The content in this file is paraphrased educational material prepared for RAG testing. Consult the linked source for the original publication, complete context, and current requirements.